List of Question of slide - 4

1. Give the basic idea of subnet (p:6)

✅ Basic Idea of Subnet

A **subnet** (short for *subnetwork*) is a logical subdivision of an IP network.

🧩 Key Concepts:

- An **IP address** consists of two main parts:
  - **Network part** – identifies the network.
  - **Host part** – identifies a specific device on that network.

➡️ Represented as: **<network> <host>**

🔧 What Subnetting Does:

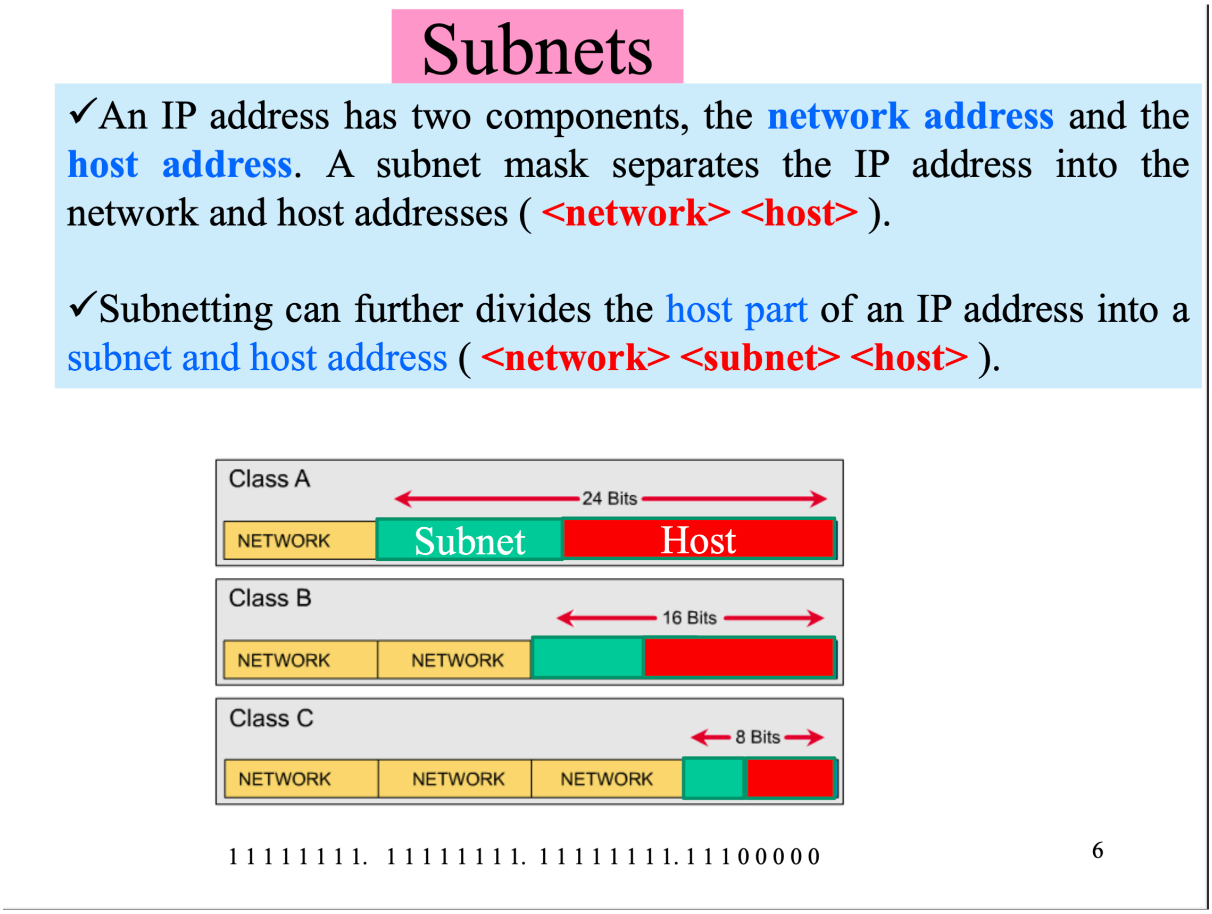
- **Subnetting** further divides the **host part** of the IP address into:
  - **Subnet** – identifies a smaller segment within the larger network.
  - **Host** – identifies individual devices within the subnet.

➡️ After subnetting: **<network> <subnet> <host>**

🧠 Why Use Subnetting?

- Improves network **organization** and **efficiency**.
- Helps in **traffic management**, **security**, and **IP address utilization**.
- Enables **creation of multiple smaller networks** within a large network.

Would you like a diagram or example to better visualize subnetting?



2. Show subnet is evaluated (p:9)

✅ How Subnet is Evaluated

To determine the **network ID** (also called **subnet**) from a given IP address, we perform a **bitwise AND operation** between the IP address and its **subnet mask**. This operation zeros out the host portion and leaves only the network part.

🧩 Example:

Let's take the IP address:

📌 **192.168.23.4** (Class C)

Default Subnet Mask for Class C:

📌 **255.255.255.0**

🔢 Step-by-Step Bitwise AND:

1️⃣ Convert to Binary:

Type	Decimal	Binary
IP Address	192.168.23.4	11000000.10101000.00010111.00000100
Subnet Mask	255.255.255.0	11111111.11111111.11111111.00000000

2️⃣ Perform Bitwise AND:

```
  11000000.10101000.00010111.00000100   (IP)
& 11111111.11111111.11111111.00000000   (Subnet Mask)
-------------------------------------------------------
  11000000.10101000.00010111.00000000   (Network ID)
```

✅ Result:

➡️ **Network ID = 192.168.23.0**

This is the **subnet address** where the host (192.168.23.4) belongs.

🧠 Key Concept:

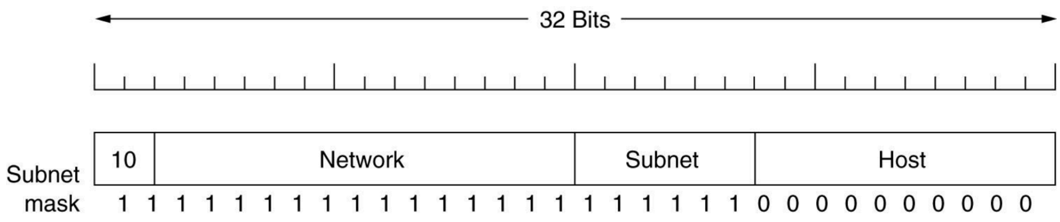
> IP Address AND Subnet Mask = Network ID

3. Problem Page no 13-15 ,20, 21, 25

- (Subtracting 2 for
network address and **broadcast address**)

Example

The example of a customized subnet mask for class B is like:



A class B network subnetted into 64 subnets.

Example: Determine the number of subnet and host per subnet

Ans. The number of subnet = $2^6 = 64$

The number of hosts/subnet = 2^{10}-2= 1022

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Example-1
If the subnet mask 255.255.240.0 is used for a class B IP address then find the number of subnets and number of hosts/subnet.

Binary: 11111111 . 11111111 . 11110000 . 00000000
Decimal: 255.255.240.0
The number of hosts/subnet = $2^{12} – 2 = 4094$
The number of subnets = $2^4 = 16$

Example-2
If the subnet mask 255.255.255.192 is used for a class C IP address then fined the number of subnets and number of hosts/subnet.

The number of hosts/subnet = $2^6 – 2 = 62$
The number of subnets = $2^2 = 4$

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Example-3

A class C IP address is 150.100.14.163 and the corresponding subnet mask is 255.255.255.128 Determine the maximum number of hosts per subnet and the number of subnets.

Ans. The subnet mask in both binary and decimal is like:

11111111. 11111111. 11111111. 10000000

255.255.255.128

Here 1 bits are for subnets and 7 bits for hosts.
Therefore, the number of hosts per subnet $2^7-2 = 126$
The number of subnets are $2^1= 2$

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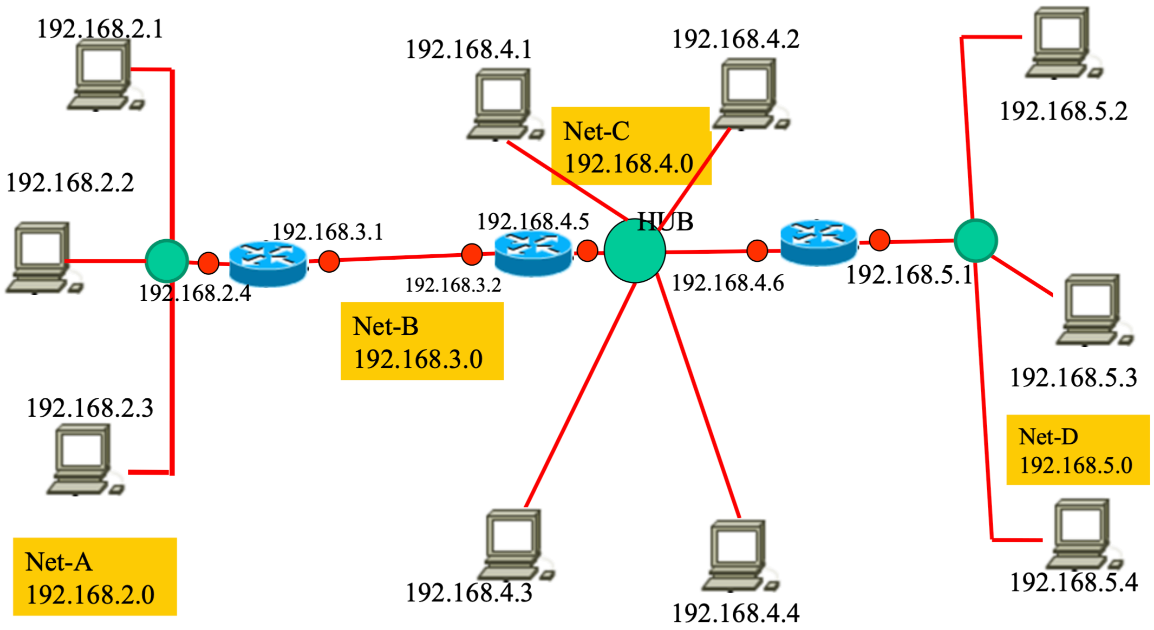
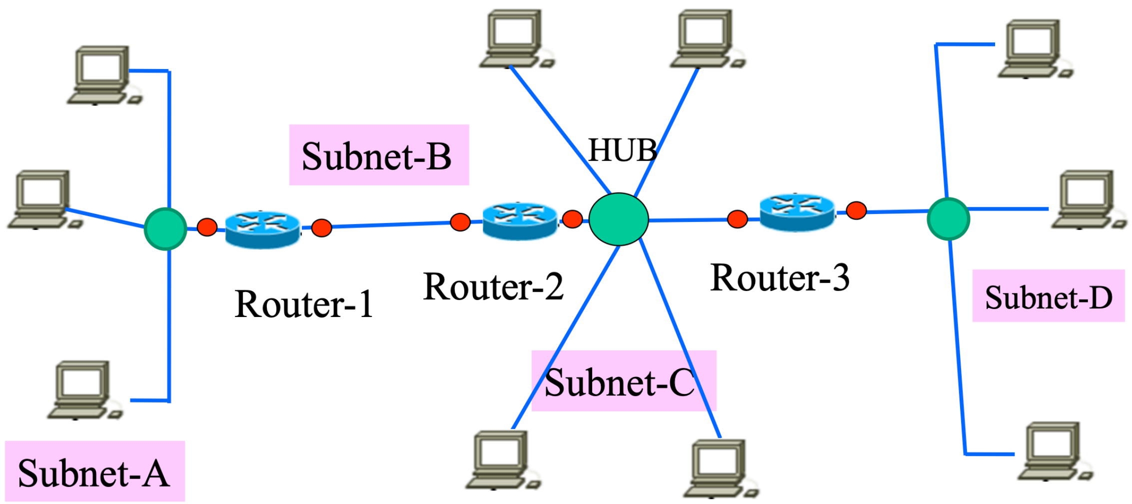
Example-1: Put default class C IP address to each node



The drawback of the network is that it needs 4 IP address from ISP for 4 networks.

Example-2 Design the network with customized subnetting using single class B IP.



We have four subnets.

a) The subnet–A has four hosts (3 PCs and 1 router interface)
b) Subnet B has two hosts correspond to Fast Ethernet ports of router-1 and router-2 which are directly connected.
c) subnet-C has 6 hosts (4 PCs and 2 router interfaces)
d) subnet-D has 4 hosts (3 PCs and 1 router interface)

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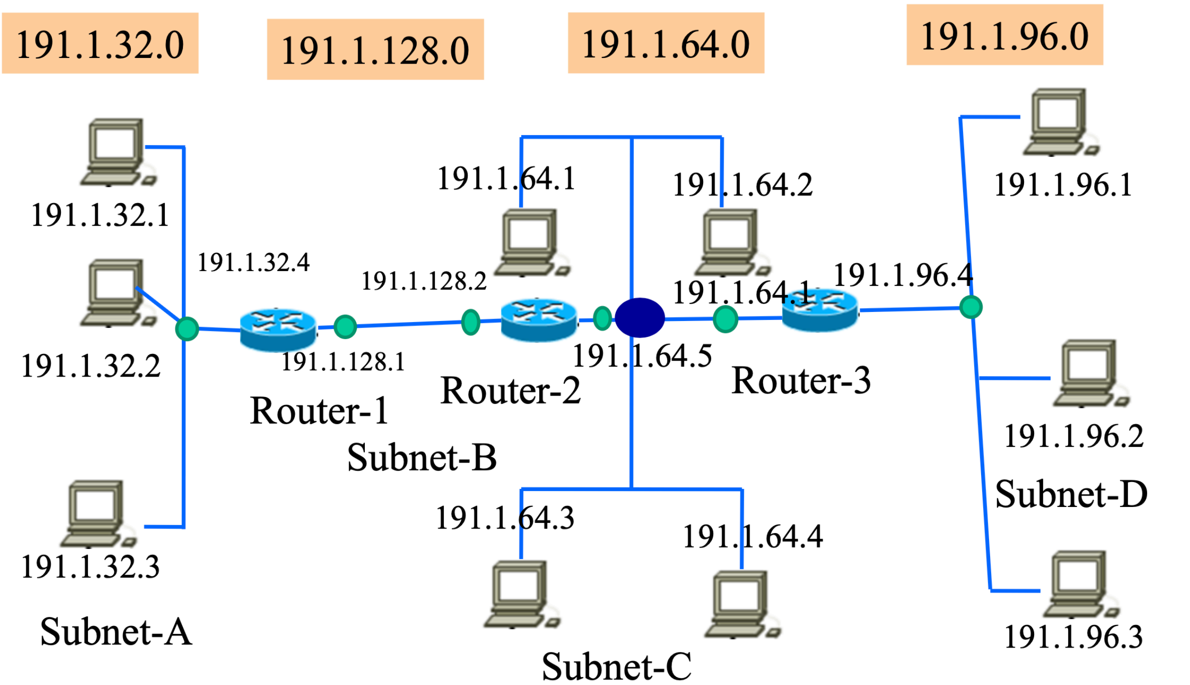

The advantage of the network is that it needs only one IP address from ISP and subnet mask of the entire network (for all the subnets) is fixed i.e. 255.255.224.0

191.1.0.0	191.1.128.0
191.1.32.0	191.1.160.0
191.1.64.0	191.1.192.0
191.1.96.0	191.1.224.0

4. What is CIDR (p:39)

💡 **CIDR (Classless Inter-Domain Routing)** is a technique used in IP addressing that allows **more flexible allocation of IP addresses** than the older **classful** system (Class A, B, C).

🔷 Key Concepts of CIDR:

1. **CIDR removes fixed class boundaries** (like Class A = /8, B = /16, C = /24) and instead uses **prefix length notation** (e.g., /24, /26).
2. CIDR allows:
    - **Subnetting** (breaking a larger network into smaller ones)
    - **Supernetting** (combining multiple smaller networks into a larger one)
3. **CIDR Notation**:

    An IP address is followed by a slash (/) and the number of bits used for the network portion.

    Example: 192.168.1.0/24

    - /24 means the **first 24 bits are for network**, and the rest for host.
4. **CIDR helps reduce IP address wastage** and improves **routing efficiency** by grouping networks into a **single route advertisement** (called route aggregation or summarization).

🔷 Example:

If you need a network with **1000 hosts**, a classful system would force you to use a **Class B** network (/16 = 65534 hosts) — wasting many IPs.

With CIDR, you can use something like /22 (1022 hosts), which is much more efficient.

Would you like an illustration showing how CIDR helps aggregate or split networks?

5. Problem on P-43

Example-8
For the starting CIDR address of 192.168.10.0/20 determine the range of class C IP address.

Ans.

Decimal	Binary of third byte
192.168.10.0	00001010
192.168.11.0	00001011
192.168.12.0	00001100
192.168.13.0	00001101
192.168.14.0	00001110
192.168.15.0	00001111

The range of class C IP address is 192.168.10.0 to 192.168.15.0

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🔍 Step-by-Step Explanation:

🧠 CIDR Basics

- The CIDR notation /20 means that the **first 20 bits** are the **network part**, and the remaining **12 bits** are for **host addresses**.
- IPv4 addresses are 32 bits long, so:

  32 - 20 = 12 bits for hosts.

🧩 IP Breakdown

Starting IP: 192.168.10.0

In binary:

- 192 → 11000000

- 168 → 10101000
- 10 → 00001010
- 0 → 00000000

So:

```
11000000.10101000.00001010.00000000 = 192.168.10.0
```

🧾 Understand the 20-bit Network

- The first 16 bits (192.168) are fixed.
- From the third byte (8 bits), only the first **4 bits** are part of the network.
- That means:
    - 3rd byte = 0000xxxx → The remaining 4 bits of the third byte and all 8 bits of the 4th byte are for hosts.

So:

- 3rd byte range = 00001010 (10) to 00001111 (15)
- That gives **192.168.10.0 to 192.168.15.255**

6. Give the concept of NAT with appropriate diagram (P: 46-47)

🧠What is NAT (Network Address Translation)?

> 💡 **NAT** is a method used by routers to translate **private IP addresses** used within a local network (LAN) into a **single public IP address** (WAN) used for communication on the internet.

It allows multiple devices on a local network to access the internet using **one public IP address**.

🔁How It Works in This Example

1. **Local Device Sends Request (Step 1):**

- A host in the LAN (10.0.0.1) sends a request to a web server on the Internet (128.119.40.186:80).
- It uses:
    - **Source IP:** 10.0.0.1
    - **Source Port:** 3345 *(chosen by host)*
    - **Destination IP:** 128.119.40.186
    - **Destination Port:** 80 (HTTP)

2. **NAT Router Translates (Step 2):**
    - The NAT router translates:
        - **Source IP** → 138.76.29.7 *(router's public IP)*
        - **Source Port** → 5001 *(random unique port chosen by router)*
    - It **stores a mapping** in the **NAT table**:

```
WAN side:   138.76.29.7:5001
LAN side:   10.0.0.1:3345
```

3. **Response from Web Server (Step 3):**
    - Web server responds to:
        - **Destination IP:** 138.76.29.7
        - **Destination Port:** 5001
    - The router checks the **NAT table** and sees:

```
138.76.29.7:5001 → 10.0.0.1:3345
```

4. **NAT Router Translates Back (Step 4):**
    - Replaces:
        - **Destination IP** → 10.0.0.1
        - **Destination Port** → 3345
    - Forwards it to the internal host.

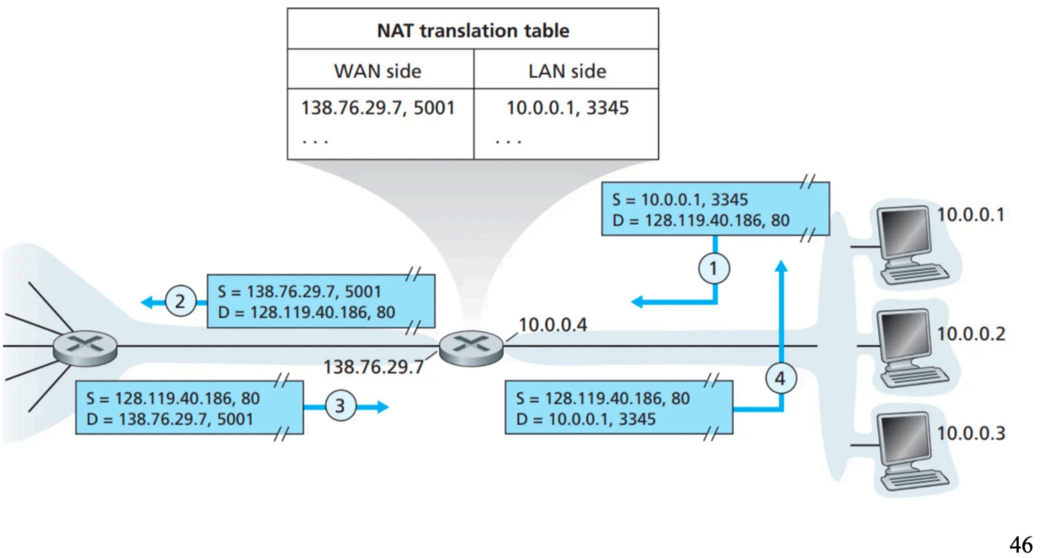
✅Key Points of NAT in the Diagram

- **Hides internal IPs**: External devices only see the public IP (138.76.29.7).
- **Uses port numbers** to distinguish between multiple internal devices.
- **Tracks mappings** of (public IP, port) ↔ (private IP, port) in a **NAT table**.
- **Enables private networks** (e.g., 10.x.x.x) to access the internet.

📌 Summary

NAT allows multiple internal devices to communicate with external networks using a single public IP address. It does so by replacing source IP and port information in outgoing packets and maintaining a mapping so that replies can be correctly routed back to the originating device inside the LAN. This process is known as **Port Address Translation (PAT)** when ports are used to distinguish individual sessions.

Would you like a simplified diagram or animation to better visualize this process?



7. How IPv6 address is represented (P : 48)

IPv6

IPv6 uses 128 bit and expressed in 32 hexadecimal numbers like:

EFAC: BA89: 7529:AFDC: 92AF:8654:1293 :29A2

After every 4 digits a colon ‘:’ is used therefore 32 digits + 7 colons = 39 characters hence called addressing system of 39 characters.

In some addresses huge number of zeros exits like:

DFAC: 0000: 0000:0000: 0009:03AC1:5923 :FEA2

can be expressed as:

DFAC: 0: 0:0: 9:3AC1:5923 :FEA2

or

DFAC: : 9:3AC1:5923 :FEA2

Double colon can be used only once in a IPv6 address.

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